

# Hardware

## Syllabus

### 3.1 Computer Architecture

- 1 (a) Understand the role of the Central Processing Unit (CPU) in a computer
- (b) Understand the role of the Central Processing Unit (CPU) in a computer
- 2 (a) Understand the purpose of the components in a CPU, in a computer that has a Von Neumann architecture
- (b) Describe the process of the fetch–decode–execute cycle including the role of each component in the process
- 3 Understand what is meant by a core, cache and clock in a CPU and explain how they can affect the performance of a CPU
- 4 Understand the purpose and use of an instruction set for a CPU
- 5 Describe the purpose and characteristics of an embedded system and identify devices in which they are commonly used

The CPU processes instructions and data that are input into the computer so that the result can be output

A microprocessor is a type of integrated circuit on a single chip

- Including:
  - **Units:** Arithmetic Logic Unit (ALU) and Control Unit (CU)
  - **Registers:** Program Counter (PC), Memory Address Register (MAR), Memory Data Register (MDR), Current Instruction Register (CIR) and Accumulator (ACC)
  - **Buses:** Address Bus, Data Bus and Control Bus
- How instructions and data are fetched from random access memory (RAM) into the CPU, how they are processed using each component and how they are then executed
- Storing data and addresses into specific registers
- Using buses to transmit data, addresses and signals
- Using units to fetch, decode and execute data and instructions
- The number of cores, size of the cache and speed of the clock can affect the performance of a CPU
- An instruction set is a list of all the commands that can be processed by a CPU and the commands are machine code
- An embedded system is used to perform a dedicated function, e.g., domestic appliances, cars, security systems, lighting systems or vending machines. This is



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different to a general-purpose computer that is used to perform many different functions, e.g., a personal computer (PC) or a laptop

### 3.2 Input & Output Devices

- 1 Understand what is meant by an input device and why it is required
  - Including:
    - Barcode Scanner
    - Digital Camera
    - Keyboard
    - Microphone
    - Optical Mouse
    - QR Code Scanner
    - Touch Screen (resistive, capacitive and infra-red)
    - Two-dimensional (2D) and three-dimensional (3D) scanners
- 2 Understand what is meant by an output device and why it is required
  - Including:
    - Actuator
    - Digital Light Processing (DLP) Projector
    - Inkjet Printer
    - Laser Printer
    - Light Emitting Diode (LED) Screen
    - Liquid Crystal Display (LCD) Projector
    - Liquid Crystal Display (LCD) Screen
    - Speaker
    - 3D printer
- 3 (a) Understand what is meant by a sensor and the purposes of sensors
  - Limited to:
    - Acoustic
    - Accelerometer
    - Flow
    - Gas
    - Humidity
    - Infra-red
    - Level
    - Light
    - Magnetic field
    - moisture
    - pH



- Pressure
- Proximity
- Temperature

(b) Identify the type of data captured by each sensor and understand when each sensor would be used, including selecting the most suitable sensor for a given context

### 3.3 Data Storage

- 1 Understand what is meant by primary storage
  - Primary storage is directly accessed by the CPU
  - Including the role of:
    - Random Access Memory (RAM)
    - Read Only Memory (ROM)
  - Including why a computer needs both RAM and ROM, and the difference between them
- 2 Understand what is meant by secondary storage
  - Secondary storage is not directly accessed by the CPU and is necessary for more permanent storage of data
- 3 Describe the operation of magnetic, optical and solid-state (flash memory) storage and give examples of each
  - Magnetic storage uses platters which are divided into tracks and sectors. Data is read and written using electromagnets
  - Optical storage uses lasers to create and read pits and lands
  - Solid-state (flash memory) uses NAND or NOR technology. Transistors are used as control gates and floating gates
- 4 Describe what is meant by virtual memory, how it is created and used and why it is necessary
  - Pages of data are transferred between RAM and virtual memory when needed
- 5 Understand what is meant by cloud storage
  - Cloud storage can be accessed remotely in comparison to storing data locally
  - Physical servers and storage are needed to store data in cloud storage
- 6 Explain the advantages and disadvantages of storing data on the cloud in comparison to storing it locally



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### 3.4 Network Hardware

- 1 Understand that a computer needs a Network Interface Card (NIC) to access a network
- 2 Understand what is meant by and the purpose of a media access control (MAC) address, including its structure
  - A network interface card is given a MAC address at the point of manufacture
  - MAC addresses are usually written as hexadecimal
  - MAC addresses are created using the manufacturer code and the serial code
- 3 (a) Understand what is meant by and the purpose of an internet protocol (IP) address
  - An IP address is allocated by the network and they can be static or dynamic  
(b) Understand that there are different types of IP address
  - Including the characteristics of and differences between IPv4 and IPv6
- 4 Describe the role of a router in a network
  - A router sends data to a specific destination on a network
  - A router can assign IP addresses
  - A router can connect a local network to the internet

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### 3.1 Computer Architecture

#### Central Processing Units

CPU (Central Processing Unit) is central to all modern computing devices. It is installed as an integrated circuit on a single microchip. It is responsible for the execution of instructions and data. It consists of:

- Control Unit (CU)
- Arithmetic & Logic Unit (ALU)
- Registers & Buses

#### Von Neumann Architecture

The Von Neumann architecture is based on the concept of a stored-program computer. A stored-program computer is a computer that stores programs and instruction in digital memory. A modern stored-program computer is one that keeps its programmed instructions, as well as its data, in read-write, random-access memory (RAM). Before a computer can actually read data, process it and produce information, it must read a set of instruction called a 'program'. The program will state what processing is required.

Four Functional Unit of Von Neumann Architecture are:

- Input/output (I/O)
- Memory
- Control Unit
- Arithmetic Logic Unit (ALU)

Von Neumann architecture had main features which are as follow:

- Concept of a central processing unit (CPU or processor)
- CPU was able to access the memory directly
- Computer memories could store programs as well as data
- Stored programs were made up of instructions which could be executed in sequential order.

Von Neumann architecture, on a simple level, state that a computer system can be through to consist of three main sections: CPU, Storage and Input/Output devices.



### Arithmetic Logic Unit

This is an internal part of the CPU that carries out calculation on data. The arithmetic part uses the usual operators such as multiply, divide, add and subtract. The logic part carries out comparisons such as 'equal to', 'greater than' and 'less than'. Values need to be placed in the accumulator for calculation to be carried out.

### Memory Unit

Computer has several types of memory. Memory unit in the Von Neumann model is the main memory, also called RAM (Random Access Memory). Main memory is used by the computer for storing a program and its data while the program is running. What distinguishes a computer from a calculator is the ability to run a stored program; main memory allows the computer to do that.

### Buses

**"The set of wires used to travel signals to and from CPU and different components of computer is called Bus."**

Bus is a group of parallel wires that is used as a communication path. As a wire transmits a single bit so 8-bits bus can transfer 8 bits (1 byte) at a time and 16-bits bus can transfer 16 bits (2 bytes) and so on. There are three types of buses according to three types of signals, these are: Data bus, Address Bus and Control Bus.

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**Data Bus:** Sends data (in both directions) between the devices (I/O), memory unit and Processor.

**Control Bus:** Carries signals (in one direction) relating to the control and coordination of all activities within the computer.

**Address Bus:** Carries signals (in one direction) relating to addresses between the memory and processor.

### Registers

Register is simply a high-speed storage area within the computer. All data must be represented in a register before it can be processed. For example, if two numbers are to be added, both numbers must be stored in registers and the result of the addition must also be stored in a register.

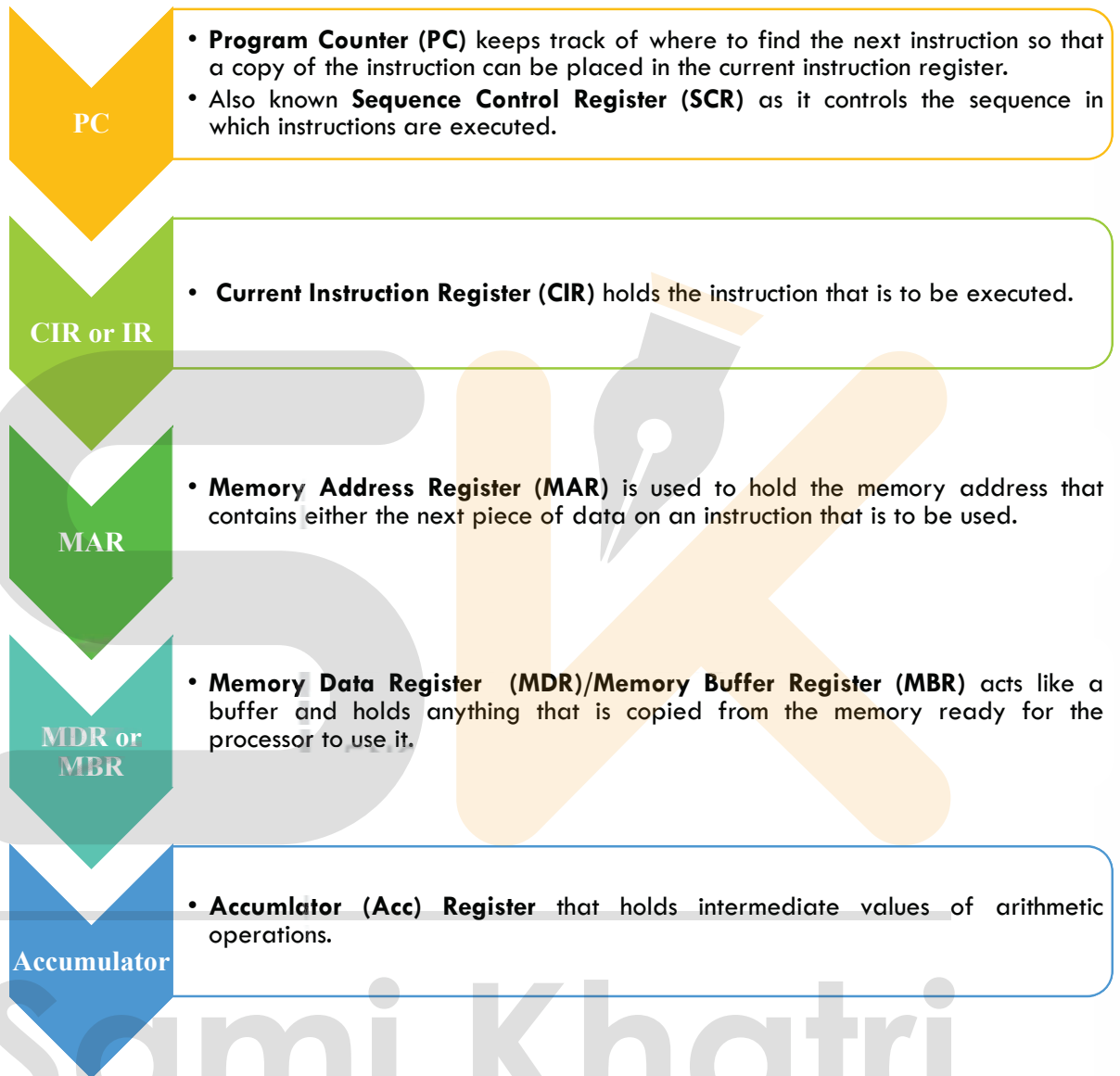


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There are five registers



### Memory

Memory is split up into many partitions, each of which contains its address (unique) and contents in 8 bits. Contents stored in binary values.

Consider two examples of how the MAR and MDR registers can be used when carrying out a read and write operation to and from memory:



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| Address   | Contents  |
|-----------|-----------|
| 1111 0000 | 0111 0010 |
| 1111 0001 | 0101 1011 |
| 1111 0010 | 1101 1101 |
| 1111 0011 | 0111 1011 |
|           |           |
| 1111 1100 | 1110 1010 |
| 1111 1101 | 1001 0101 |
| 1111 1110 | 1000 0010 |
| 1111 1111 | 0101 0100 |

Consider the **READ** operation. Suppose we want to read the contents of memory location 1111 0001; the two registers are used as follows:

Address location 1111 0001 to be read from first written into the MAR (Memory Address Register):

MAR: 1 1 1 1 0 0 0 1

'Read Signal' is sent to the computer memory » the contents of memory location 1111 0001 are then put into the MDR (memory data register):

MDR: 1 0 0 1 0 1 0 1

Consider the **WRITE** operation. To show how the value 1001 0101 was written into memory location 1111 1101:

Data to be stored is first written into the MDR (Memory Data Register):

MDR: 1 0 0 1 0 1 0 1

Data has to be written into location with address: 1111 1101; so this address is now written into the MAR:

MAR: 1 1 1 1 1 1 0 1

'Write Signal' is sent to the computer memory and the value 10010101 will then be written into the correct memory location.

### Input/output Devices

The input and output devices are discussed in Input and Output Devices Section and are the main method of entering data into and getting data out of computer systems. Input devices convert external data into a form the computer can understand and can then process (e.g. keyboards, touchscreens and microphones). Output devices show the results of computer processing in a humanly understandable form (e.g. printers, monitors and loudspeakers).



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## Control Unit

**Control Unit (CU)** controls the operation of the memory, processor and input/output devices. It contains the **Current Instruction Register (CIR)** and the **Program Counter (PC)**. The CIR contains the current instruction during processing. The PC contains the address of the next instruction to be executed, the control unit reads an instruction from memory (the address of the location where the instruction can be found is stored in the Program Counter (PC)).

**System Clock** is used to produce timing signals on the control bus to ensure this vital synchronization takes place - without the clock the computer would simply crash.

The RAM is often referred to as the **Immediate Access Store (IAS)**. The CPU takes data and programs held in backing store (e.g. a hard disk drive) and puts them into RAM temporarily.

## Clock Speed

It determines the number of cycles the CPU can execute per second. Increasing clock speed increases the number of operations/numbers of fetch-execute cycles that can be carried out per unit of time, however, there is a limit on clock speed.

## Fetch Decode Execute Cycle

To carry out a set of instructions, the processor first of all fetches some data and instructions from memory and stores them in suitable registers. Both the address bus and the data bus are used in this process. Once this is done, each instruction needs to be decoded before finally being executed. This is all known as the Fetch-Decode-Execute Cycle and is the last part of this puzzle.

**Fetch:** Next instruction, which is currently stored in memory address in MAR and instruction in MDR, is fetched. Contents of the MDR are then copied to the Current Instruction Register (CIR). The PC is then incremented (increased by 1) so that the next instruction can be then be processed.

**Decode:** The instruction is then decoded so that it can be interpreted in the next part of the cycle.

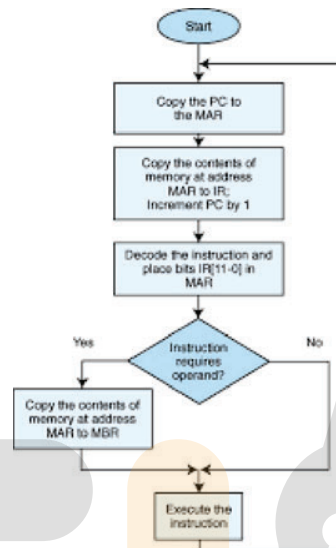
**Execute:** The CPU passes the decoded instruction as a set of control signals to the appropriate components within the computer system. This allows each instruction to be carried out in its logical sequence.



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## Cache

Most data is stored on the hard disk. When we use that data, it is loaded into Random Access Memory (RAM) because it is much faster to access the data from RAM than from the hard disk.

When we looked at the 'Immediate Access' for earlier, we found out that the CPU needs to access data very quickly. Although RAM is faster than the hard disk, it still isn't fast enough to cope with the speed that the CPU needs to deal with data.

To appreciate this, imagine a race taking place between two cars, one is a powerful high speed sports car and the other is an ordinary little run-about. Their task is to race across town but there are many traffic lights in the way. The small car goes down and the big car hoots off the line and the little car just toots along, well behind. Moments later, the big car has to stop at a traffic light, so within a minute the small car catches up once again. By

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the time the race is over, the big car is hardly in front of the slow car because of all the traffic light delays. This is exactly what happens in a CPU. The processor may sit around idly waiting for new data to arrive and so it appears less powerful than it really is.

The way around this, is to store as much data as possible, as close as possible to the CPU to avoid delays. This nearby storage area is called the 'cache'. The cache is a special type of computer memory which can be accessed much faster than RAM. The CPU looks in the cache for the data it needs. If the data is there, it will retrieve it and process it. If the data is not there, then the CPU accesses the system memory and then puts a copy of the new data in the cache before processing it. Cache memory is volatile i.e. when the computer is shut down, the data stored there is lost. Thus, in order to improve the computer's performance, it is important to keep the local cache full at all times so that the CPU doesn't have to wait around with nothing to do.

The larger the cache size, the better the performance because you are able to store more data locally.

#### Factors

- Performance and speed increases with increasing bus.
- Increase in clock speed with increase the speed.
- Performance can be changed by altering bus width, clock speed and use of multi-core CPU's.
- Use of cache memories can also speed up a CPU's performance.

#### Instruction Set

Set of Operations which are decoded in a particular sequence. Each operation will instruct the ALU and CU (which are part of the CPU). An operation is made up of an opcode (Operation needs to be done) and an operand (data which needs to be acted on or it can refer to a register in the memory).

Computer needs to understand the operation to be carried out, there is actually a limited number of opcodes that can be used; this is known as the Instruction Set.

One example of an instruction set is the X86. a common CPU standard used in „ many modern computers.

Instruction Sets are the low-level language instructions that instruct the CPU how to carry out an operation. Program code needs interpreters or compilers to convert the code into the instruction set understood by the computer.

#### Operating System

The software that controls the allocation and usage of hardware resources such as memory, central processing unit (CPU) time, disk space, and peripheral devices. The operating system

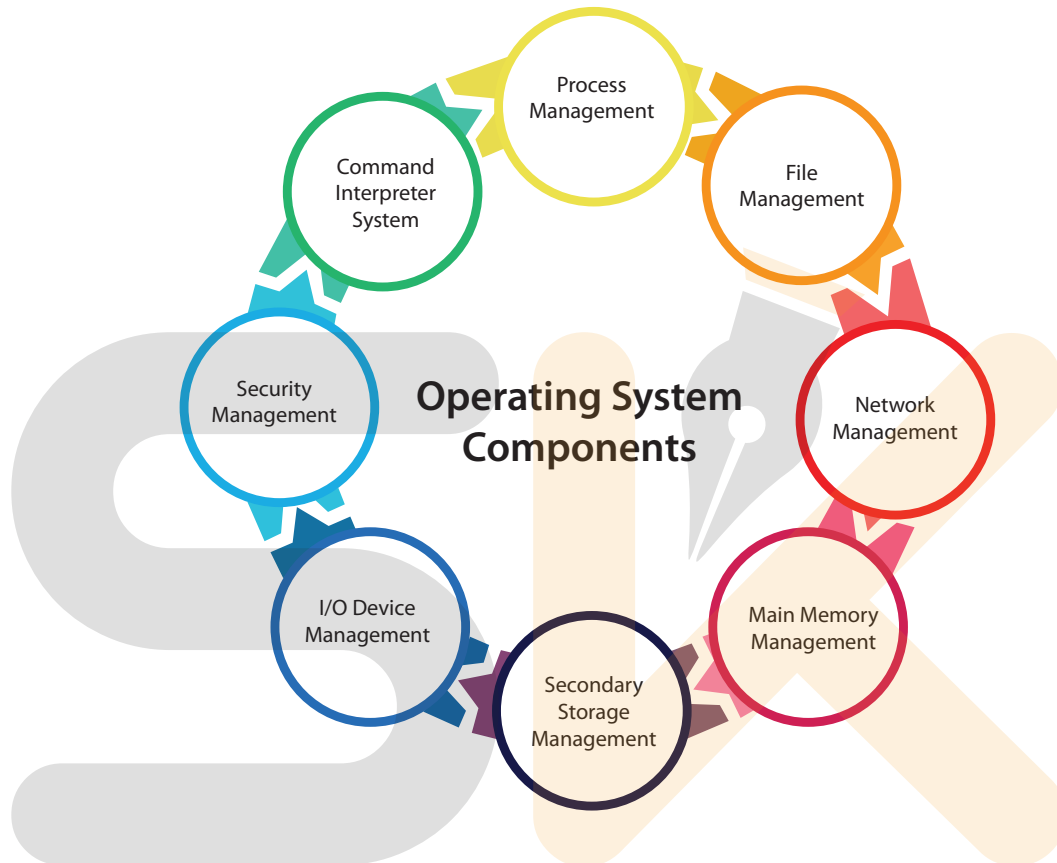


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is the foundation software on which applications depend. Popular operating systems include Windows 98, Windows NT, Mac OS, and UNIX. Acronym: OS.



### Functions of Operating System

- Accepting data from input devices and transferring it to the computer's memory.
- Making sure that any output is sent to the correct output device.
- It manages the transfer of data between the computer's memory and backing storage devices.
- Allocating memory space to programs and data.
- Keeping track of which parts of the memory have already been allocated and the parts that are still free.
- It deals with the loading of applications software into memory and controls the execution, or 'running' of them.
- It also provides a way for applications software to communicate with the computer's hardware.
- It deals with any errors that occur when a program is being run, or when data is being transferred somewhere, and informs the user if necessary.
- Monitoring and restricting access to programs and data.



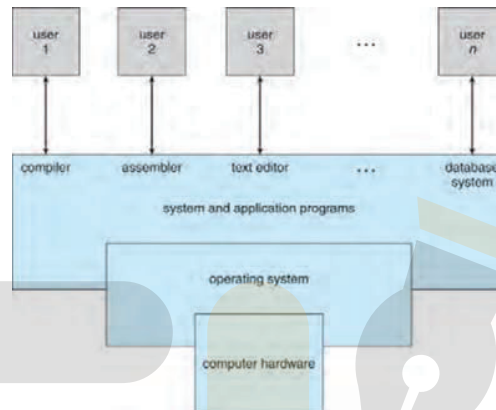
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- It provides a human computer interface, or HCI, for the user.
- Preventing unauthorized access to the system.

Operating System acts as bridge between user and applications.



## Interrupt

An interrupt is a signal to the processor emitted by hardware or software indicating an event that needs immediate attention. An interrupt alerts the processor to a high-priority condition requiring the interruption of the current code the processor is executing. The processor responds by suspending its current activities, saving its state, and executing a function called an interrupt handler (or an interrupt service routine, ISR) to deal with the event. This interruption is temporary, and, after the interrupt handler finishes, the processor resumes normal activities.

## Types of Interrupts

**Hardware Interrupts** are used by devices to communicate that they require attention from the operating system.

**Software Interrupt** is caused either by an exceptional condition in the processor itself, or a special instruction in the instruction set which causes an interrupt when it is executed.

## Buffers

Temporary storage/memory that holds data being transferred between devices often used to compensate for different speeds of devices examples printer, disk, etc.

## Embedded System

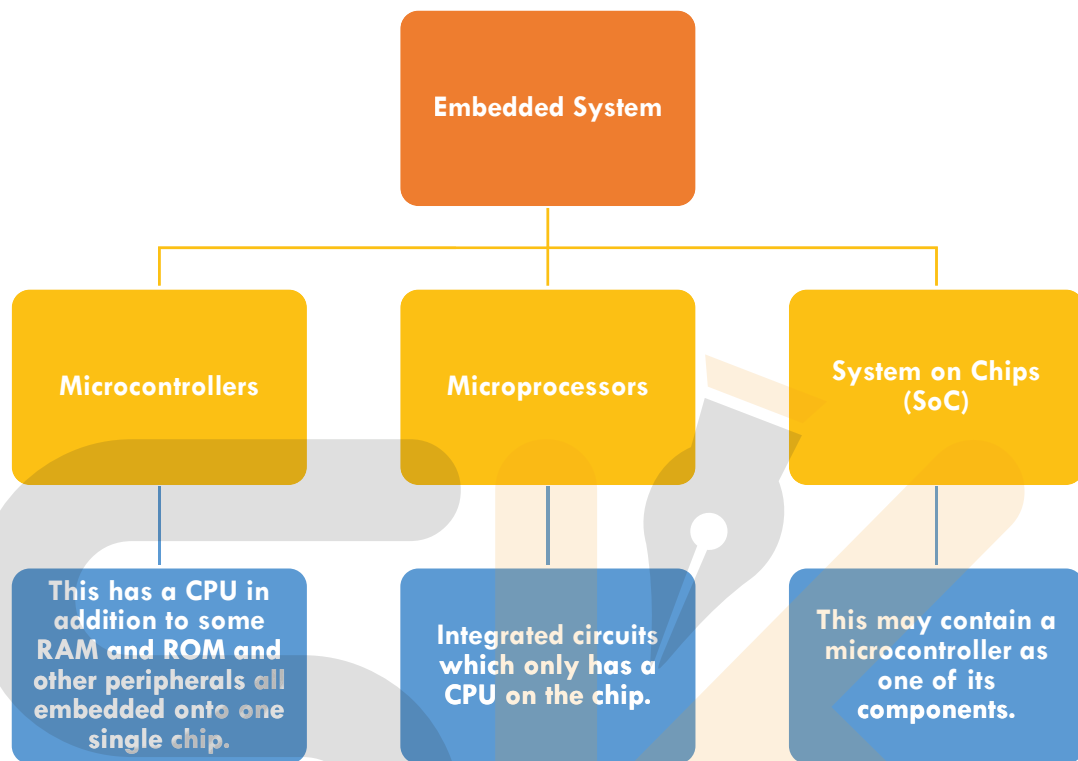
Embedded System is a combination of hardware and software which is designed to carry out a specific set of functions. The hardware is electronic, electrical or electro-mechanical. Embedded systems can be based on:



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Embedded System will have a specific set of tasks:

Depending on the device, embedded systems are either programmable or nonprogrammable. Non-programmable devices need, in general, to be replaced if they require a software upgrade. Programmable devices permit upgrading by two methods:

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- ➡ Connecting the device to a computer and allowing the download of updates to the software.
- ➡ Automatic updates via a Wi-Fi, satellite or cellular (mobile phone network) link.

**Pros & Cons**

| Pros                                                 | Cons                                       |
|------------------------------------------------------|--------------------------------------------|
| Small in size and therefore easy to fit into devices | Difficult to upgrade                       |
| Cheap                                                | Troubleshooting fault                      |
| One task, simple interface, no Operating System      | Confusing interface (appears to be simple) |
| Consume very little power                            | Can be hacked                              |
| Can be controlled remotely                           | Difficult to repair                        |
| Fast reaction                                        | Wasteful                                   |
| Mass production comes reliability                    |                                            |

**Example 1: Motor Vehicles**

Modern cars have many parts that rely on embedded systems to function correctly. Figure shows some of the many components that are controlled in this way.

**Example 2: Set top Box**

Set-top box uses an embedded system to allow, for example, recording and playback of television programs. This can be operated remotely by the user when not at home using an internet-enabled device or by using the interface panel when at home.

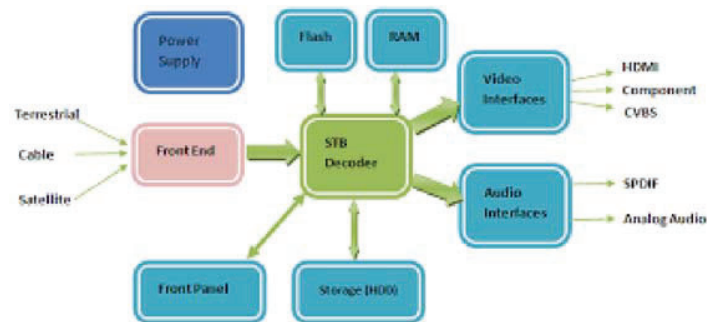
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### Example 3: Security System

The security code is set in RAM and the alarm activated or deactivated using the keypad. Data from sensors is sent to the controller which checks against values stored on the SSD.

An output can be a signal to flash lights, sound an alarm or send a message to the home owner via their mobile phone. Again, the home owner can interface with the system remotely if necessary.

### Example 4: Vending Machines

Vending machines make considerable use of embedded systems. They usually use microcontrollers to control a number of functions that we all associate with vending machines:

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## 3.2 Input & Output Devices

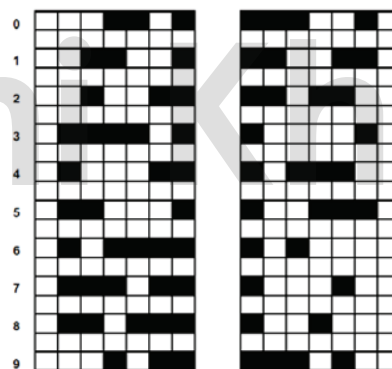
### Input Devices

A barcode is simply a numeric code represented as a series of lines. These lines can be read by a barcode reader/scanner. The most common use of barcode readers is at Point-ofSale (POS) in a shop. The code for each item to be purchased needs to be entered into the computer. Reading the barcode is far quicker and more accurate than typing in each code using a keypad. Barcode can be found on many other items that have numeric codes which have to be read quickly and accurately - for example ID

### Barcode Scanners

A barcode is a series of dark and light parallel lines of varying thickness. The numbers 0 to 9 are each represented by a unique series of lines. Numerous barcode methods for representing these digits exist. The example we shall use adopts different codes for digits appearing on the left and for digits appearing on the right of the barcode:

Each digit in the barcode is represented by bars of 1 to 4 blocks thick as shown in figure Note there are different patterns for digits on the left-hand side and for digits on the right-hand side.



### Process

- ➡ It is scanned using barcode scanner through red laser.
- ➡ Light is reflected back off in the barcode.



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- Reflected light is read by sensors.
- Pattern is generated which is converted into digital data.

After reading barcode

- Search the barcode against the item in the inventory.
- Item details are sent back to checkout.
- Item is removed from the inventory and inventory is updated.

#### **Advantages for Management**

- Easier & faster to alter prices.
- give instantaneous and comprehensive sales trends.
- No need to price each item (saves time and money).
- Allows automatic stock control.
- Check customer's buying habits more easily.

#### **Advantages for Customers**

- Faster checkouts
- Less chance of errors
- Get an itemized bill
- Cost savings can be passed on to customer
- Better tracking of 'sell by dates'

It is used in many stores, libraries, supermarkets etc.

#### **Quick Response (QR) Codes**

QR codes are a type of barcode. However, they can hold considerably more data. QR codes are usually read by built-in cameras in smart phones or tablets using an installed app. Once the QR code has been scanned, it will send information back to the phone or tablet, e.g. a website link or general information (e.g. a phone number or 'special offer').

#### **Pros of QR Codes**

- There is no need for the user to write down or key in a website address; scanning the QR code does this automatically
- QR codes can store website addresses/URLs that appear in magazines, trains, buses or even on business cards, giving a very effective method of advertising.
- Data is stored within the pattern, so no need for a connected database
- Simple and quick way to store and retrieve information



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- Can store a variety of information

### Cons of QR Codes

- More than one format.
- It can transmit malicious codes.

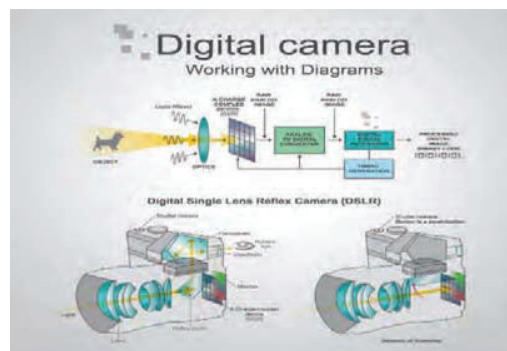
### Digital Cameras

Digital cameras have essentially replaced the more traditional camera that used film to capture the images. The film required developing and then printing before the photographer could see the result of their work.

Digital cameras contain a microprocessor which automatically:

- adjusts shutter speed
- focuses
- operates the flash
- adjusts the aperture
- removes 'red eye'
- reduces hand shake, and so on

The images recorded on a digital camera are stored on memory cards, although some may have a limited amount of external memory of its own. Digital cameras feature an LCD screen which allow you to preview and review your images, plus change menu settings. Modern digital cameras simply link to a computer system via a USB port or by using Bluetooth (which enables wireless transfer of photographic files).



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## Keyboards

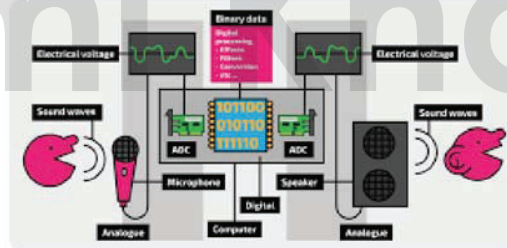
Keyboards are the most common input device; keys are pressed by the operator to enter data directly into the computer. When a key is pressed, it completes a circuit and a signal is sent to the microprocessor which interprets which key has been pressed. Since entering data by keyboard is a relatively slow process, most computer systems use a keyboard buffer which prevents the microprocessor waiting for keys to be pressed. Although little training is needed to use a keyboard, they do pose a health risk to operators who do a lot of continuous typing – the health risk is known as RSI (repetitive strain injury).



## Microphone

Microphones are used to input sound into a computer. When the microphone picks up sound, a diaphragm vibrates producing an electric signal. A sound card in the computer converts the signal into digital values which can be stored in its memory.

The electric current output from the microphone can also be sent to a computer where a sound card converts the current into a digital signal which can then be stored in the computer. The following diagram shows what happens when the word 'hut' is picked up by a microphone and is converted into digital values:



## Optical Mouse

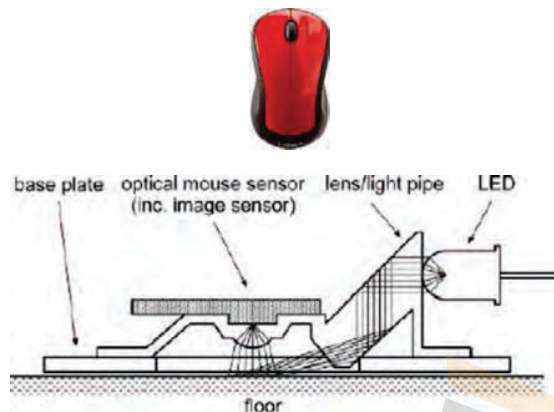
An optical mouse is an example of a pointing device. It uses tiny cameras to take 1500 images per second. Unlike an older mechanical mouse, the optical mouse can work on virtually any surface.



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## Scanners

In computer science, an image scanner—often abbreviated to just scanner. Scanner is a device that optically scans images, printed text, handwriting, or an object, and converts it to a digital image.

### 2D-Scanner

These types of scanner are the most common form and are generally used to input hard-copy (paper) documents. The image is converted into an electronic form which can be stored in a computer. A number of stages occur when scanning a document:



Cover is raised

- Document is placed on glass panel.
- Cover is closed.

Bright light then illuminates the document

- Use a type of xenon lamp or LED that produces a very bright light.

Scan head moves across the document

- Lens focuses the documented image.

Focused image now falls onto a CCD

- Each element creates an electric charge when light falls on it.
- Software produces a digital image from the electronic form.



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They come in two varieties, flatbed or handheld. Some printers have inbuilt flatbed scanners. The functionality of 2D scanners is enhanced when combined with OMR and OCR software. For example, OCR software can convert scanned in text into a computer readable form.

### 3D-Scanners

3D scanners scan solid objects and produce an electronic 3D image. They make use of, for example, lasers, magnetic resonance, white light or X-rays. Many different technologies can be used to build these 3D-scanning devices; each technology comes with its own limitations, advantages and costs.

CT (computed tomography) scanners are used to produce a 3D image of a solid Object. Tomography technology splits up the object into a number of very thin 'Slices'. They tend to use X-rays, radio waves or gamma imaging methods. The resultant image allows a solid object to be stored as a series of digital Values representing each 'slice'.

- **X-rays:** CT (computed tomography) scanners.
- **Radio waves:** MRI (magnetic resonance imaging) scanners.
- **Gamma rays:** SPECT (single-photon emission computed tomography) scanners.

### lication

COMPUTED TOMOGRAPHIC (CT) SCANNERS are used to create a 3D image of a solid object. This is based on TOMOGRAPHY technology which basically builds up an image of the solid object through a series of very thin 'slices'. Together these 3D 'slices' make up a representation of the 3D solid object.

Each slice is built up by use of X-rays, radio frequencies or gamma imaging; although a number of other methods exist. Each 'slice' is then stored as a digital image in the computer memory. The whole of the solid object is represented digitally in the computer memory. Depending on how the image is formed, the type of tomographic scanner can have different names.

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## Touch Screens

Touchscreens are now a very common form of input device. They allow simple touch to launch an application or to carry out many of the functions of pointing devices such as a mouse. One of the main uses of touchscreen technology is in mobile phones. At present, there are three major types of touchscreen technologies applied to mobile phone screens:

- ➡ Capacitive
- ➡ Infrared
- ➡ Resistive

## Sensors

Sensors are input devices that record data about the physical environment around it. Sensors send data to a microprocessor (computer). They do not make judgements, decisions or control any output devices. There are many types of sensors used in a variety of household, commercial and industrial applications.

| Sensor Types        | Use and Features of Sensors                                                                                                         |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Temperature         | Measure the (indoor) temperature [°C]<br>Average range of values : -25 to +7(C°                                                     |
| Humidity            | Measure the (indoor) humidity [%]<br>Average range of values : 0 to 100%                                                            |
| Lighting            | Measure the (indoor) illumination [lx]<br>Average range of values : 200 to 1900lx                                                   |
| Energy Consumption  | System meters/power plug meters. System for temporal tracking of energy consumption for per item [kWh]                              |
| Occupancy/ Movement | Systems for crowd counting through WIFI, GSM, Bluetooth signals or through volume recognition with depth sensors [Number of People] |
| Air Quality         | Measure of CO <sub>2</sub> and TVOC [ppm] in the indoor environment<br>Average range of values : 0 to 2000ppm                       |
| Noise Level         | Measure the intensity of sound [dB] in the indoor environment.                                                                      |

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## Output Devices

### Actuators

When a computer is used to control devices, such as a conveyer belt or a valve, it is usually necessary to use an actuator to, for example, start/stop the conveyer belt or open/close the valve. An actuator is a mechanical or electromechanical device such as a relay, solenoid or motor. We will consider a solenoid as the example; this converts an electrical signal into a magnetic field producing linear motion:



### Light Projectors

Projectors are used to project computer output onto larger screens or even onto interactive whiteboards. They are often used in presentations and in multimedia applications. The next section compares the basic operation of the two projector technologies.



There are two types of Light Projectors



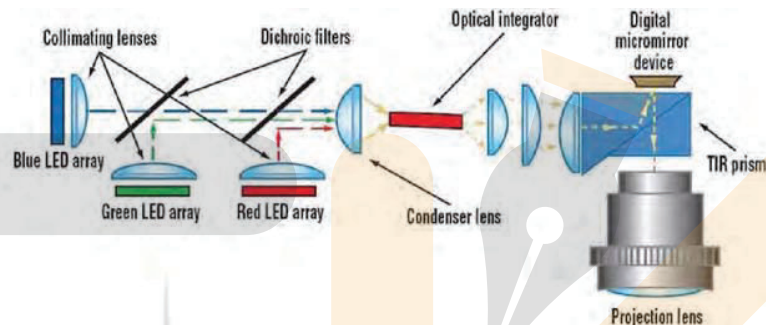
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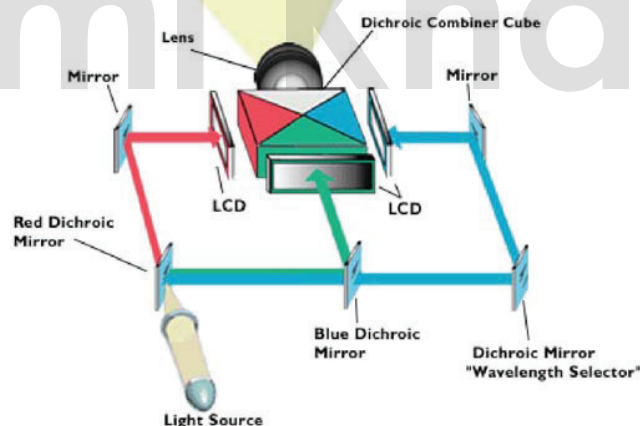
### Digital Light Projectors (DLP)

DLP (Digital Light Processing) is a proprietary technology developed by Texas Instruments. It works quite differently than LCD. Instead of having glass panels through which light is passed, the DLP chip is a reflective surface made up of thousands of tiny mirrors. Each mirror represents a single pixel. In a DLP projector, light from the projector's lamp is directed onto the surface of the DLP chip. The mirrors wobble back and forth, directing light either into the lens path to turn the pixel on, or away from the lens path to turn it off.



### Liquid Crystal Display (LCD) Projector

- Older technology than DLP
- A powerful beam of white light is generated from a bulb
- This beam of light is then sent to a group of chromatic-coated mirrors; these reflect the light back at different wavelengths
- When the white light hits the mirrors, the reflected light has wavelengths corresponding to red, green and blue
- These three different lights pass through three LCD screens; these screens show the image to be projected as millions of pixels in grayscale
- When the colored light passes through the LCD screens, a red, green and blue version of the grey image emerges
- Finally, the image passes through the projector lens onto the screen



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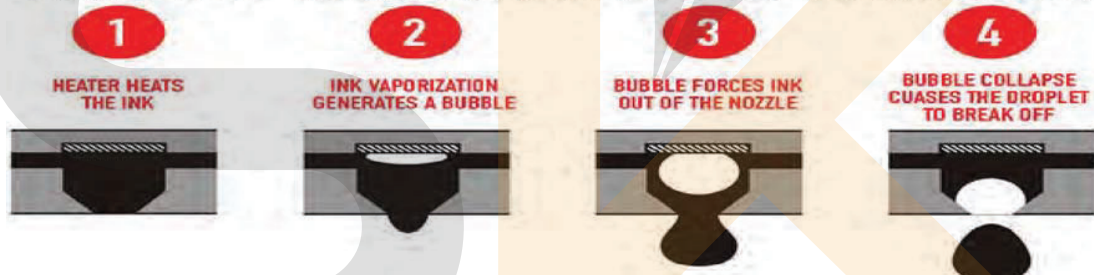
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## Inkjet Printers

INKJET PRINTERS are essentially made up of:

- Print head which consists of nozzles which spray droplets of ink on to the paper to form characters
- Ink cartridge or cartridges; either a group of cartridges for each colour (blue, yellow and magenta) and a black cartridge or one single cartridge containing all three colours + black (Note: some systems use six colors.)
- Stepper motor and belt which moves the print head assembly across the page from side to side
- Paper feed which automatically feeds the printer with pages as they are required.

### 4 STEPS OF THERMAL INKJET (TIJ) TECHNOLOGY



## Laser Printers

LASER PRINTERS differ greatly from inkjet printers in the way they print pages. They use dry powder ink rather than liquid ink and make use of the properties of static electricity to

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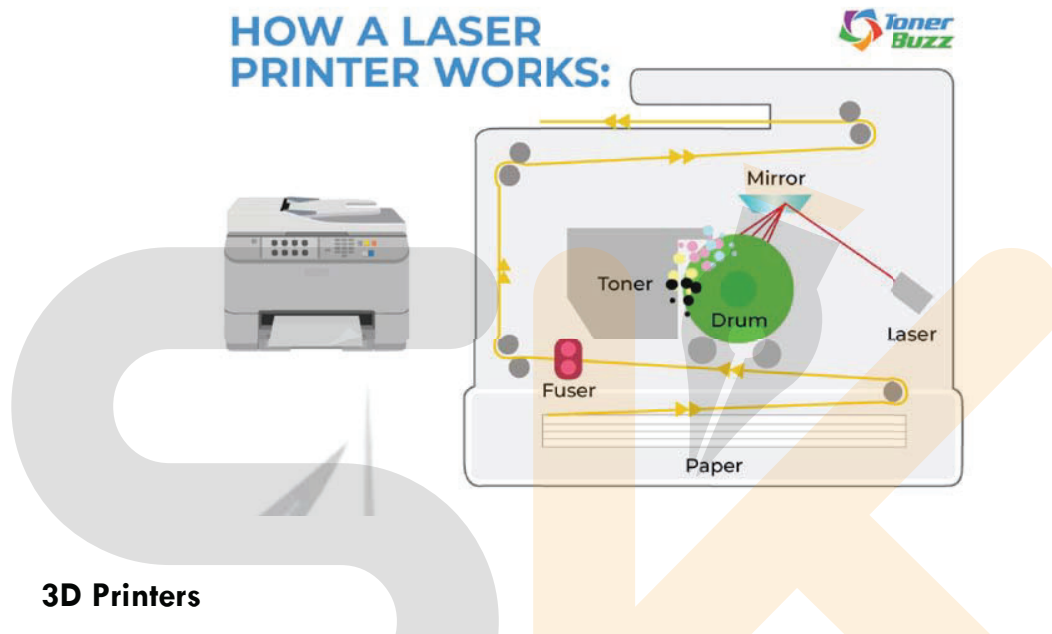


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produce the text and images. Unlike inkjet printers, laser printers print the whole page in one go (inkjet printers print the page line by line). Their advantage is the speed at which they can carry out large print jobs (e.g. 2000 leaflets) and the fact that they don't run out of ink halfway through.



### 3D Printers

3D printing, also known as additive manufacturing (AM), refers to processes used to create a three-dimensional object in which layers of material are formed under computer control to create an object. Objects can be of almost any shape or geometry and typically are produced using digital model data from a 3D model or another electronic data source.

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The term "3D printing" originally referred to a process that deposits a binder material onto a powder bed with printer heads layer by layer. More recently, the term is being used in popular vernacular to encompass a wider variety of additive manufacturing techniques. United States and global technical standards use the official term additive manufacturing for this broader sense.

### Uses

3D printing is regarded as being possibly the next 'industrial revolution' since it will change the manufacturing methods in many industries. The following list is just a glimpse into what we know can be made using these printers; in the years to come, this list will probably fill an entire book:

- Prosthetic limbs made to exactly fit the recipient
- Items to allow precision reconstructive surgery (e.g. facial reconstruction following an accident); the parts made by this technique are more precise in their design since they are made from exact scanning of the skull
- In aerospace, manufacturers are looking at making wings and other parts using 3D technology; the bonus will be lightweight precision parts
- In fashion and art – 3D printing allows new creative ideas to be developed
- Making parts for items no longer in production, e.g. suspension parts for a vintage car.

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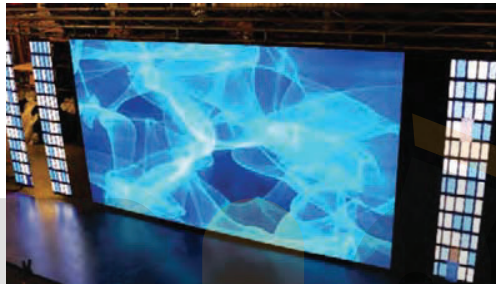
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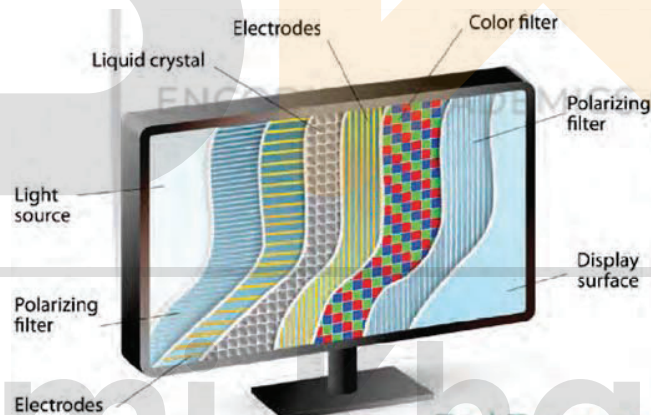
## LED Screens

An LED screen is made up of tiny light emitting diodes (LEDs). Each LED is either red, green or blue in color. By varying the electric current sent to each LED, its brightness can be controlled, producing a vast range of colors. This type of screen tends to be used for large outdoor displays due to the brilliance of the colors produced.



## LCD Screens

LCD screens are made up of tiny liquid crystals. These tiny crystals make up an array of pixels that are affected by changes in applied electric fields. How this works is outside the scope of this book. But the important thing to realize is that for LCD screens to work, they require some form of backlighting.



## OLED (Organic Light Emitting Diode)

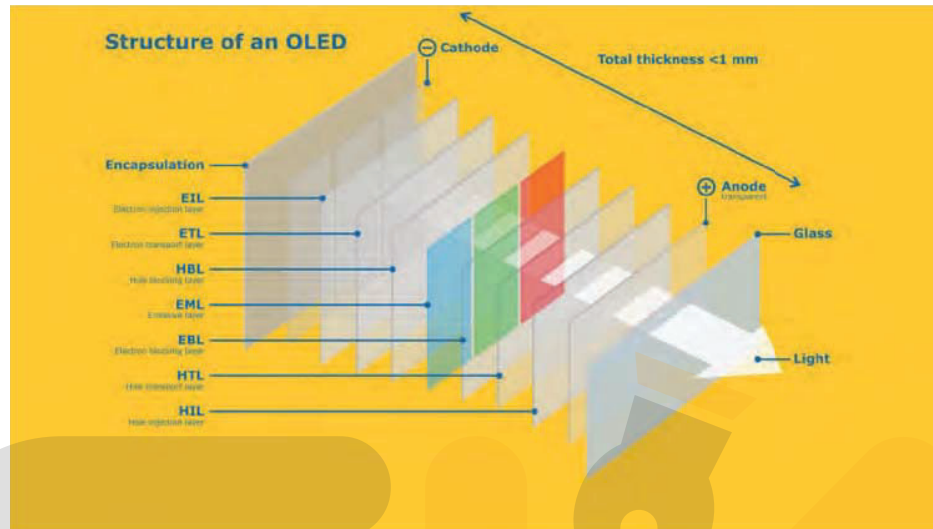
Future LED technology is making use of ORGANIC LIGHT EMITTING DIODES (OLEDs). These use organic materials (made up of carbon compounds) to create semi-conductors which are very flexible. Organic films are sandwiched between two charged electrodes (one is a metallic CATHODE and the other a glass ANODE). When an electric field is applied to the electrodes they give off light. This means that no form of backlighting is required. This allows for very thin screens. It also means that there is no longer a need to use LCD technology, since OLED is a self-contained system.



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### Advantages

- Plastic, organic layers of an OLED are thinner, lighter and more flexible than the crystal structures used in LEDs or LCDs
- Light-emitting layers of an OLED are lighter; OLED layers can be made from plastic rather than the glass used in LED and LCD screens
- OLEDs give a brighter light than LEDs
- OLEDs do not require backlighting like LCD screens – OLEDs generate their own light
- OLEDs require no backlighting, they use much less power than LCD screens (most of the LCD power is used to do the backlighting); this is very important in battery-operated devices such as mobile phones
- OLEDs are essentially plastics, they can be made into large, thin sheets (this means they could be used on large advertising boards in airports, subways, and so on)
- OLEDs have a very large field of view, about 170 degrees, which makes them ideal for use in television sets and for advertising screens.



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## Loudspeakers

Loudspeakers are output devices that produce sound. When connected to a computer system, digitized sound stored on a file needs to be converted into sound as follows:

- The digital data is first passed through a digital to analogue converter (DAC) where it is changed into an electric current.
- This is then passed through an amplifier (since the current generated by DAC will be very small); this creates a current large enough to drive a loudspeaker.
- This electric current is then fed to a loudspeaker where it is converted into sound.

The following schematic shown:

## Monitoring Applications

Monitoring System will carry out the following applications:

- System is activated by a password on a keypad
- The infrared sensor will pick up the movement of an intruder in the building
- The acoustic sensor will pick up sounds such as footsteps or breaking glass
- The pressure sensor will pick up the weight of an intruder coming through a door or through a window
- Sensor data is passed through an ADC if it is in an analogue form to produce digital data
- The computer/microprocessor will sample the digital data coming from these sensors at a given frequency (e.g. every 5 seconds)
- The data is compared with the stored values by the computer/microprocessor
- If any of the incoming data values are outside the acceptable range, then the computer sends a signal to a siren to sound the alarm, or to a light to start flashing
- DAC is used if the devices need analogue values to operate them
- Alarm continues to sound/lights continue to flash until the system is reset with a password

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### Patient Monitoring

### Control Applications

### Street Lights

Microprocessor is used to control the operation of a street lamp. The lamp is fitted with a light sensor which constantly sends data to the microprocessor. The data value from the sensor changes according to whether it is sunny, cloudy, raining or it is night time.



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### **Anti-lock Braking System**

Anti-lock braking systems (ABS) on cars use magnetic field sensors to stop the wheels locking up on the car if the brakes have been applied too sharply.



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### 3.3 Data Storage

#### Data Storage

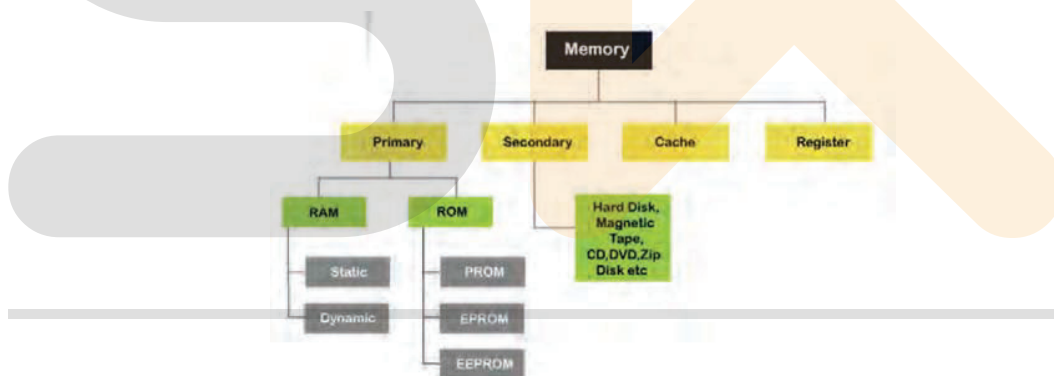
Computers require some form of memory and storage. Memory is usually referred to as the internal devices used to store data that the computer can access directly. This is also known as primary memory. This memory can be the user's workspace, temporary data or data that is key to running the computer.

Storage devices allow users to store applications, data and files. The user's data is stored permanently and they can change it or read it as they wish.

Storage devices can also be removable to allow data, for example, to be transferred between computers. Removable devices allow a user to store important data in a different location in case of data loss.

It can be split into two groups:

- Primary Memory
- Secondary Memory



A computer holds programs and data in three sorts of device:

- **Primary:** Limited capacity and rapid-access processing.
- **Secondary:** Larger capacity and slower access to keep data/programs for future use. They remain inside computer.
- **Off-line:** Portable, they are removed after read and write.



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## Primary Memory

Primary Storage, also referred to as computer memory, is used by the CPU (Central Processing Unit) of a computer to continuously read and execute programming instructions. Also it reads and writes data from/to the file store in the Secondary Storage of the computer. And this is accessible directly by the CPU. Typical examples of Primary Storage types are RAM, ROM, and memory adapters.

### RAM

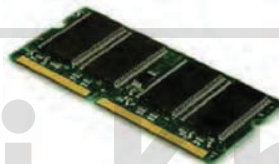
This is a volatile memory (i.e. contents are lost when the computer is switched off). A user can write or delete data as well as read the contents. Before a program can be ran it must first be loaded into RAM which allow the CPU to access the program. Battery-backed RAMs overcome data loss on switching power off.

RAM never runs out of memory; it continues to operate but just becomes slower and slower as more data is stored.

There are two types of RAM's:

#### DRAM

Dynamic random-access memory (DRAM) is a type of storage that is widely used as the main memory for a computer system. Dynamic random-access memory (DRAM) is a type of random-access memory that stores each bit of data in a separate capacitor within an integrated circuit. The capacitor can be either charged or discharged; these two states are taken to represent the two values of a bit, conventionally called 0 and 1. A DRAM storage cell is dynamic in that it needs to be refreshed or given a new electronic charge every few milliseconds to compensate for charge leaks from the capacitor.



#### SRAM

SRAM (static RAM) is random access memory (RAM) that retains data bits in its memory as long as power is being supplied. Unlike dynamic RAM (DRAM), which stores bits in cells consisting of a capacitor and a transistor, SRAM does not have to be periodically refreshed.



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| SRAM vs. DRAM                                  |                                                            |
|------------------------------------------------|------------------------------------------------------------|
| STATIC RAM                                     | DYNAMIC RAM                                                |
| Does not require refreshing                    | Must be continuously refreshed                             |
| Requires multiple transistors to store one bit | Requires one transistor and one capacitor to store one bit |
| Delivers faster access times                   | Delivers slower performance                                |
| Consumes less power, especially in idle        | Consumes more power                                        |
| Takes up more space                            | Requires less space                                        |
| Can hold only a small amount of data           | Can hold much more data                                    |
| Costs more than DRAM                           | Costs less than SRAM                                       |
| Typically used for a processor's cache         | Typically used for a computer's main memory                |

### Applications of RAM

- ➔ Program in own routines; new instructions stored in RAM chip.
- ➔ RAM chip will store the data/instructions received from the remote control unit.

### ROM

The ROM contains the program that is used to start up the computer. This is also called the boot program because the process of starting up the computer is also called 'booting up' the computer. In many desktop computer systems, the boot program is also called the BIOS or Basic Input Output System. ROM is not volatile and is the same even when the computer is turned off. Also, the program inside the ROM is not easily deleted or changed. This means the boot program will always be there when the computer is started. This is why any computer system needs some ROM. On a standard personal computer, the boot program does the following things:

- ➔ It performs some basic checks.
- ➔ It finds the operating system and loads it into the RAM.
- ➔ It then hands control over to the operating system.

### Applications of ROM

- ➔ Storing the factory settings such as remote control frequencies
- ➔ Storing the 'start-up' routines when the toy car is first switched on
- ➔ Storing of the set routines; for example, how the buttons on the hand-held device control turning left, acceleration, stopping, and so on.

| RAM                      | VS | ROM                 |
|--------------------------|----|---------------------|
| Random Access Memory     |    | Read Only Memory    |
| RAM is volatile.         |    | ROM is non-volatile |
| Allows Reading & Writing |    | Allows Reading Only |
| Temporary Storage        |    | Permanent Storage   |
| RAM is expensive         |    | ROM is cheap        |
| Performs R&W functions   |    | Performs R function |
| DRAM & SRAM              |    | PROM & EPROM        |



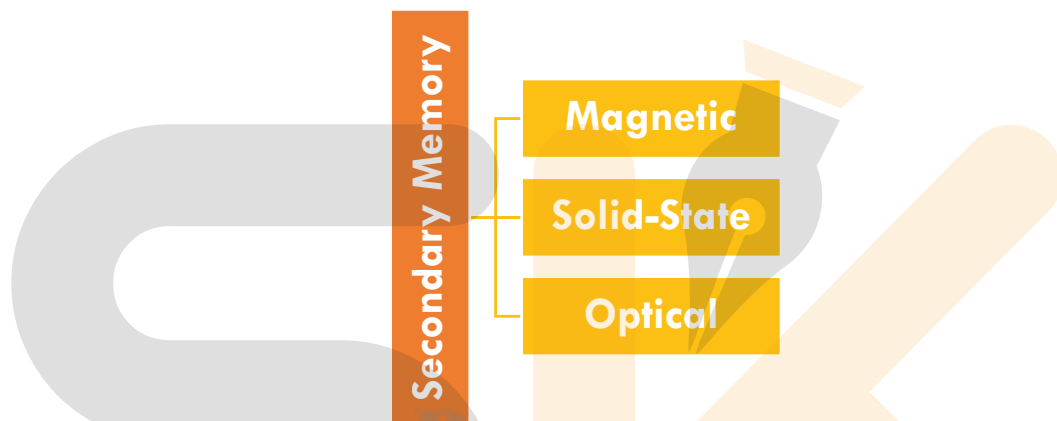
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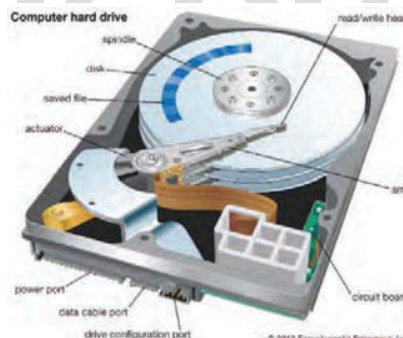
## Secondary Memory

These storage devices are external to the basic built in of the computer processor and are used to build more space for storage. They are non-volatile devices that allow data to be stored as long as required by the user. This type of storage can store more data than primary memory, but data access time is considerably longer than with RAM or ROM. The secondary storage cannot be changed, but the information sustains even if the computer is turned off.



### Magnetic (Hard Drive)

Hard-drives have a very large storage capacity (up to 10 TB). They can be used to store vast amounts of data. Hard-drives are random access devices and can be used to store all types of files, including huge files such as movies. Data access speeds are very fast. Data is stored inside a hard-drive on rotating metal or glass discs (platters). Hard disk stores information in the form of magnetic fields. Data is stored digitally in the form of tiny magnetized regions on the platter where each region represents a bit. To write a data on the hard disk, a magnetic field is placed on the tiny field in one of these two polarities: N-S – If North Pole arrives before the South Pole and S-N – if the South Pole arrives before the North Pole while the field is accessed. An orientation in the one direction (like NS) can represent the '1' while the opposite orientation (S-N) represents '0'. This polarity is sensed by integrated controllers built within the hard disk.



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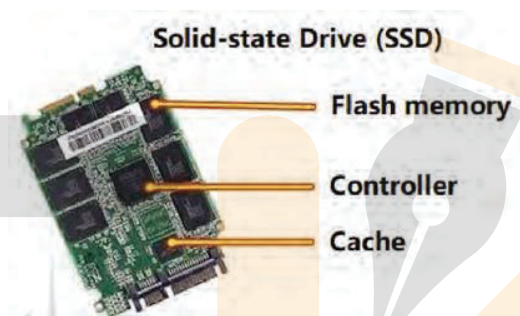


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Removable hard disk drives are essentially HDDs external to the computer that can be connected to the computer using one of the USB ports. In this way, they can be used as a back-up device or another way of transferring files between computers.

### **Solid State (SSD)**

The term solid-state essentially means no moving parts. Solid-state storage devices are based on electronic circuits with no moving parts (no reels of tape, no spinning discs, no laser beams, etc.)



The traditional spinning hard drive (HDD) is the basic nonvolatile storage on a computer. That is, it doesn't "go away" like the data on the system memory when you turn the system off. Hard drives are essentially metal platters with a magnetic coating. That coating stores your data, whether that data consists of weather reports from the last century, a high definition copy of the Star Wars trilogy, or your digital music collection. A read/write head on an arm accesses the data while the platters are spinning in a hard drive enclosure.

An SSD does much the same job functionally (saving your data while the system is off, booting your system, etc.) as an HDD, but instead of a magnetic coating on top of platters, the data is stored on interconnected flash memory chips that retain the data even when there's no power present. The chips can either be permanently installed on the system's motherboard (like on some small laptops and ultra-books), on a PCI/PCI e card (in some high-end workstations), or in a box that's sized, shaped, and wired to slot in for a laptop or desktop's hard drive (common on everything else). These flash memory chips differ from the flash memory in USB thumb drives in the type and speed of the memory. That's the subject of a totally separate technical treatise, but suffice it to say that the flash memory in SSDs is faster and more reliable than the flash memory in USB thumb drives. SSDs are consequently more expensive than USB thumb drives for the same capacities.

Memory sticks/flash memories (also known as pen drives) use solid state technology. They usually connect to the computer through the USB port. Their main advantage is that they are very small, lightweight devices, which make them very suitable as a method for transferring files between computers.



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## Optical Storage

Optical storage devices save data as patterns of dots that can be read using light. A laser beam is the usual light source. The data on the storage medium is read by bouncing the laser beam off the surface of the medium. If the beam hits a dot it is reflected back differently to how it would be if there was no dot. This difference can be detected, so the data can be read. Dots can be created using the laser beam (for media that is writable such as CD-Rs). The beam is used in a high-power mode to actually mark the surface of the medium, making a dot. This process is known as burning data onto a disc. This is a magnified view of the dots on the surface of a CD. The different patterns of dots correspond to the data stored on the disc. A CD has digital information stored in the form of laser-created pits on its surface. These in turn can be read by detecting the laser light scattered from the pit. Stores data as tiny pits pressed into flat surface by laser. Optical discs such as CDs and DVDs have smaller capacities than HDDs or tapes and are also more expensive per gigabyte of storage. They are also direct access media, with rather slower access than a HDD, and are removable and highly portable.

Blu-ray discs are another example of optical storage media. However, they are fundamentally different to DVDs in their construction and in the way they carry out read-write operations.

Since Blu-ray discs can come in single layer or dual-layer format (unlike DVD, which is always dual-layer), it is probably worth also comparing the differences in capacity and interactivity of the two technologies.

| PARAMETER          | DVD                                                                                                                                                                                                                            | BLUE-RAY DISC                                                                                                                                                                                                                                                                             |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STORAGE CAPACITY   | <ul style="list-style-type: none"> <li>Single layer DVDs(DVD-5s) can store about 4.7 GB.</li> <li>Double layer DVDs(DVD-9s) can store about 8.7 GB.</li> </ul>                                                                 | <ul style="list-style-type: none"> <li>Single layer Blu-ray discs store approximately 25 GB.</li> <li>Double layer Blu-ray discs can store about 50 GB.</li> </ul>                                                                                                                        |
| LASER TECHNOLOGY   | <ul style="list-style-type: none"> <li>Use a red laser of longer wavelength i.e. 650nm wave length to read DVD discs.</li> <li>Red lasers are wider in diameter and thus the reading is comparatively less precise.</li> </ul> | <ul style="list-style-type: none"> <li>Use a blue laser of shorter wavelength i.e. 405nm to read the stored information.</li> <li>Blue lasers are two and a half times smaller in diameter than red lasers and thus allows for closer and more precise reading of information.</li> </ul> |
| DISC CONSTRUCTION  | Grooves on its underside are to be made wide enough to accommodate the larger wavelength. (resulting in lesser storage)                                                                                                        | Grooves on a Blu-ray disc are much thinner and closer together because the blue laser used to read the disc has a shorter wave length.(resulting in almost 5 times more grooves and immense storage)                                                                                      |
| SCRATCH RESISTANCE | It has a protective layer(0.6 mm) to resist scratching.                                                                                                                                                                        | It has a physically thinner(0.1 mm) layer, but with a hard coating that makes it more scratch resistant.                                                                                                                                                                                  |
| IMAGE RESOLUTION   | <ul style="list-style-type: none"> <li>Standard definition resolution of 480p</li> <li>Enhanced definition resolution of 520p</li> </ul>                                                                                       | High definition resolution of 1080p                                                                                                                                                                                                                                                       |
| DATA TRANSFER RATE | <ul style="list-style-type: none"> <li>Data 11.08 Mbps</li> <li>Audio/Video 10.08 Mbps</li> </ul>                                                                                                                              | <ul style="list-style-type: none"> <li>Data 36.0 Mbps</li> <li>Audio/Video 54.0 Mbps</li> </ul>                                                                                                                                                                                           |
| VIDEO RESOLUTION   | 720x480 (480i/480p-US)                                                                                                                                                                                                         | 1920x1080 (1080p)                                                                                                                                                                                                                                                                         |
| AUDIO CODECS       | <ul style="list-style-type: none"> <li>Linear PCM</li> <li>Dolby Digital</li> <li>DTS Digital Surround</li> </ul>                                                                                                              | <ul style="list-style-type: none"> <li>Linear PCM</li> <li>Dolby Digital</li> <li>DTS Digital Surround</li> <li>Dolby Digital Plus</li> <li>Dolby TrueHD</li> <li>DTS-HD</li> </ul>                                                                                                       |
| VIDEO CODECS       | <ul style="list-style-type: none"> <li>MPEG-2</li> </ul>                                                                                                                                                                       | <ul style="list-style-type: none"> <li>MPEG-2</li> <li>MPEG-4 AC</li> <li>SMPTE VC-1</li> </ul>                                                                                                                                                                                           |



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The most common use of DVD and Blu-ray is the supply of movies or games. The memory capacity of CDs isn't big enough to store most movies.

## Virtual Memory

If the amount of available RAM is exceeded due to multiple programs running, it is likely to cause a system crash. This can be solved by utilizing the hard disk drive (or SSD) if we need more memory. This is the basis behind virtual memory. Essentially RAM is the physical memory, while virtual memory is RAM + swap space on the hard disk or SSD.

The main benefits of virtual memory are:

- Programs can be larger than physical memory and still be executed
- No need to waste memory with data that isn't being used (e.g. during error handling)
- Reduces the need to buy and install more expensive RAM memory

**Drawback:** When using HDD for virtual memory the main drawback is disk thrashing. As main memory fills, more and more data needs to be swapped in and out of virtual memory leading to a very high rate of hard disk read/write head movements, this is known as disk thrashing.

## Cloud Storage

Cloud storage is a method of data storage where data is stored on remote servers. The same data is stored on more than one server in case of maintenance or repair, allowing clients to access data at any time. This is known as data redundancy. The physical environment is owned and managed by a hosting company and may include hundreds of servers in many locations.

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### Types of Cloud Storages

**Public Cloud:** Storage environment where the customer/client and cloud storage provider are different companies.



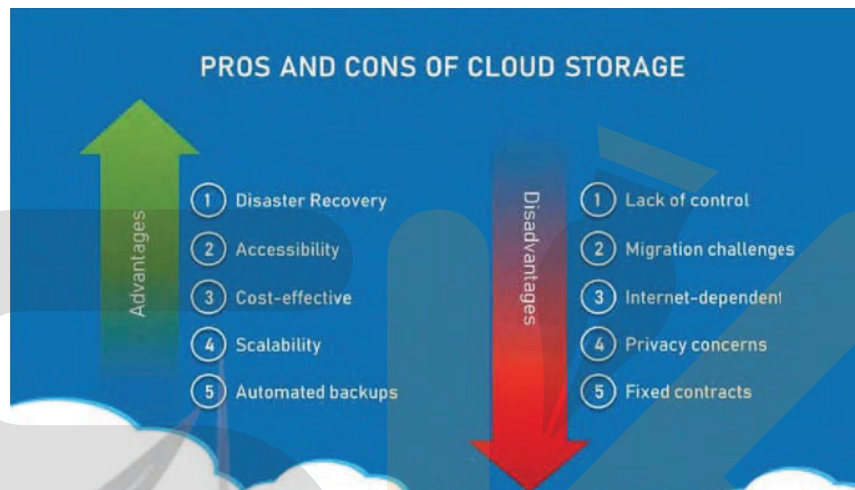
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**Private Cloud:** computing services offered either over the Internet or a private internal network and only to select users instead of the general public.

**Hybrid Cloud:** Combination of the two above environments, some data resides in the private cloud and less sensitive/less commercial data can be accessed from a public cloud storage provider.



### Data Security

Companies that transfer vast amounts of confidential data from their own systems to a cloud service provider are effectively relinquishing control of their own data security.

There is a risk that important and irreplaceable data could be lost from the cloud storage facilities. Actions from hackers (gaining access to accounts or phishing attacks, for example) could lead to loss or corruption of data. Users need to be certain that sufficient safeguards exist to overcome these risks.

## 3.4 Network Hardware

### Network Interface Card (NIC)

A network interface card (NIC) is needed to allow a device to connect to a network (such as the Internet). It is usually part of the device hardware and contains the Media Access Control (MAC) address generated at the manufacturing stage.

While the standard NIC is a plastic circuit board that slides into a computer to connect with the motherboard, there are multiple ways this connection can occur:

- Wireless
- Wired
- USB
- Fiber optics



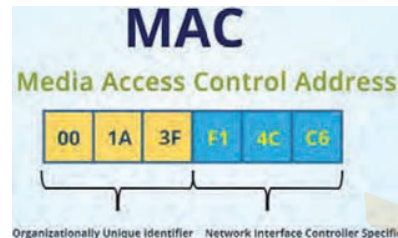
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## Media Access Control (MAC)

A MAC address is made up of 48 bits which are shown as six groups of hexadecimal digits with the general format:



For example, 00 - 1C - B3 - 4F - 25 - FF where the first six hex digits identify the device as made by, for example, HP and the second set of six hex digits are the serial number of the device itself (this is unique). If the NIC card is replaced, the MAC address will also change.

### Types of MAC Addresses

It should finally be pointed out that there are two types of MAC address: Universally Administered MAC Address (UAA) and the Locally Administered MAC Address (LAA).

The UAA is by far the most common type of MAC address and this is the one set by the manufacturer at the factory. It is rare for a user to want to change this MAC address.

There are a few reasons why the MAC address needs to be changed using LAA:

- Certain software used on mainframe systems need all the MAC addresses of devices to fall into a strict format.
- It may be necessary to bypass a MAC address filter on a router or a firewall.
- To get past certain types of network restrictions it may be necessary to emulate unrestricted MAC addresses.

## Internet Protocol (IP) Address

An IP address is a unique address that identifies a device on the internet or a local network. Set of rules governing the format of data sent via internet or local network.

There are different two versions of IP i.e. IPv4 and IPv6.

- IPv4 is based on 32-bits and the address is written as four groups of eight bits. For example: 254.15.28.77
- IPv6 is based on 128-bits address and it take the form of eight groups of hexadecimal digits. For example A8FB: 7A88:FF00:0FFF:3D21:2085:66FB:F0FA.  
IPv6 has been designed to allow the internet to grow in terms of the number of hosts and potential increase in the amount of data traffic.



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The main advantages of IPv6 compared to IPv4 are

- Removes the risk of IP address collisions
- Has built-in authentication checks
- Allows for more efficient packet routes.

| IPv4                                                       | IPv6                                                                                    |
|------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Address Size:<br>32-bit number                             | Address Size:<br>128-bit number                                                         |
| Address Format:<br>Dotted Decimal Notation:<br>192.168.1.1 | Address Format:<br>Hexadecimal Notation:<br>fe80:94db:946e:8d4e:129e                    |
| Prefix Notation:<br>255.255.255.0<br>/24                   | Prefix Notation:<br>/64                                                                 |
| Number of addresses:<br>$2^{32} = 4,294,967,296$           | Number of addresses:<br>$2^{128} = 340,282,366,920,938,463,463,374,607,431,768,211,456$ |

IP addresses can be either static (don't change) or dynamic (change every time a device connects to the internet).

#### Static IP Address

Static IP addresses are permanently assigned to a device by the internet service provider (ISP); they don't change each time a device logs onto the internet.

#### Dynamic IP Address

Dynamic IP addresses are assigned by the ISP each time a device logs onto the internet. This is done using Dynamic Host Configuration Protocol (DHCP). A computer on the internet configured as a DHCP server, is used by the ISP to automatically assign an IP address to a device.

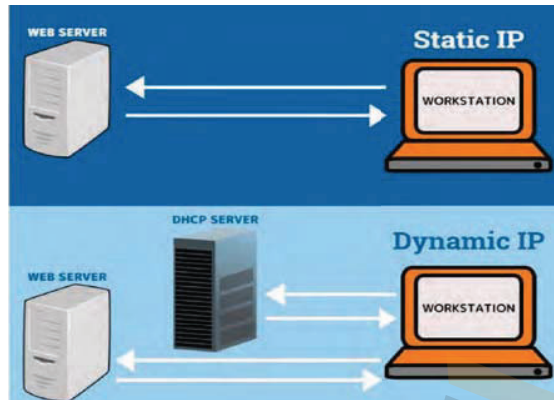
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## STATIC IP ADDRESS VERSUS DYNAMIC IP ADDRESS

### STATIC IP ADDRESS

A permanent numeric address manually assigned to a device in the network

Assigned manually by the network administrator

Does not change once it is assigned to a device

Less secure

Assigning is difficult

Suitable for dedicated services such as mail, FTP and VPN servers

### DYNAMIC IP ADDRESS

A temporary IP address that is assigned to a device or a node when it is connected to a network

Assigned by the DHCP server automatically

Changes each time the device connects to the network

More Secure

Assigning is easier

Suitable for a large network that requires internet access to all devices



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## Routers

Routers enable data packets to be routed between different networks, for example, to join a LAN to a WAN. The router takes data transmitted in one format from a network (which is using a particular protocol) and converts the data to a protocol and format understood by another network, thereby allowing them to communicate. A router would typically have an internet cable plugged into it and several cables connecting to computers and other devices on the LAN.



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